Introduction to Artificial Intelligence (AI)

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Course Outline

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2. Major Historical Milestones
3. AI Taxonomy
4. Examples of Tasks
5. Main Limitations
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8. Sources and exercice are on Github NicolasCaronPro/Introduction-to-AI-courses

What is AI?

Audience question

What is, according to you, artificial intelligence?

A Working Definition of AI

Artificial Intelligence

Artificial Intelligence is the field that studies how to design machines capable of performing tasks that usually require human intelligence.

Examples of such abilities include:

- perception,
- reasoning,
- learning,
- planning,
- decision-making,
- interaction with humans or environments.

Major Historical Milestones

II. Major Historical Milestones

Audience question

In what year do you think artificial intelligence started?

What was the latest major AI revolution? Why?

Historical Milestones

Period	Milestone	Main idea
1940s	Alan Turing and Enigma	Computation and cryptanalysis
1950	Turing Test	Can machines imitate intelligence?
1956	Dartmouth workshop	Birth of AI as a research field
1980s	Expert systems	Rule-based reasoning
2012	AlexNet / ImageNet	Deep learning breakthrough
2017	Transformer	Attention-based architectures
2022+	Generative AI / LLMs	Large-scale language models

Sources: Turing (1950), McCarthy et al. (1955), Krizhevsky et al. (2012), Vaswani et al. (2017).

Example: Alan Turing and Enigma

Historical example

During World War II, Alan Turing contributed to cryptanalysis work related to the Enigma machine. This is not "AI" in the modern machine-learning sense, but it is an important step in the history of computation.

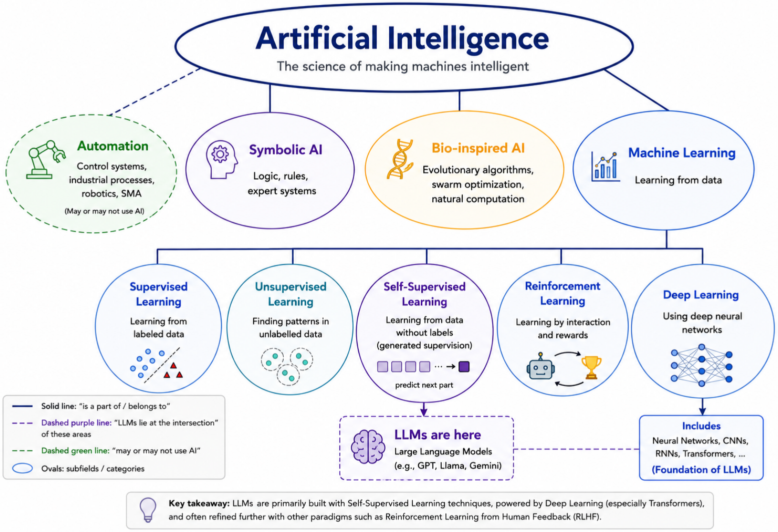
Why is it relevant?

- It illustrates that intelligence-related tasks can be formalized.
- It connects reasoning, computation, and machines.
- Turing later proposed the famous imitation game, now known as the Turing Test.

Source: Turing's 1950 paper is widely recognized as foundational in discussions on machine intelligence.

AI Taxonomy

Main Categories of AI



Key message

Key message

Artificial Intelligence is broader than Machine Learning: it also includes automation, symbolic reasoning, biologically-inspired systems, and (for some people) multi-agent decision systems (not the same thing as LLM multi-agent).

Examples of Tasks

IV. Examples of Tasks

Audience question

List some examples of the use of AI.

Classification: Cat or Dog?

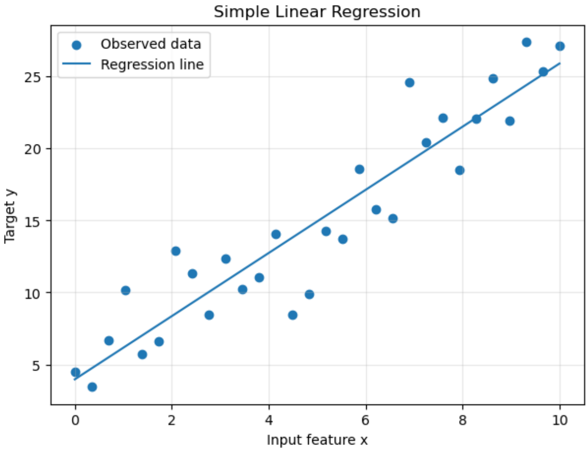
Task

Given an input image, predict a discrete class.

$$f_\theta(x) \to \{\text{cat}, \text{dog}\}$$

- Output: class label
- Example: image recognition
- Learning type: supervised learning

Regression: Predicting a Continuous Value



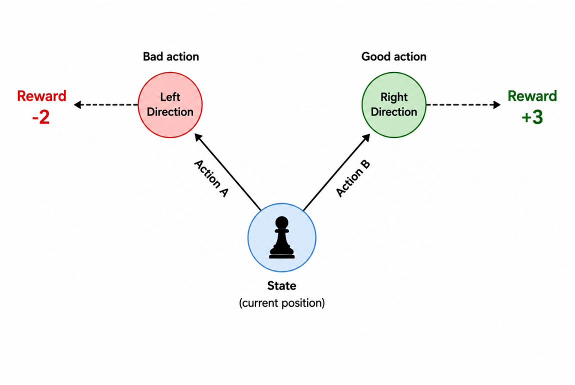
Task

Given input variables, predict a continuous value.

$$f_\theta(x) \to y \in \mathbb{R}$$

- Output: numerical value
- Example: price, temperature, demand
- Core idea: fit a function

Game AI



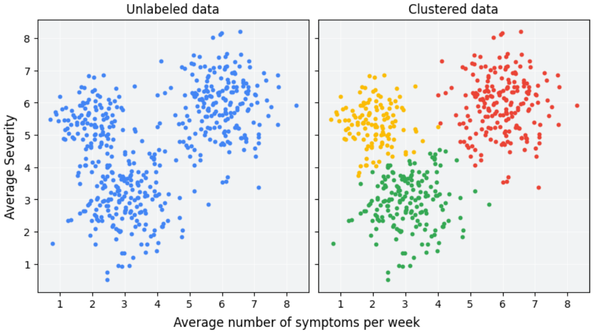
Task

Given a game state, choose the next action.

$$f_\theta(s) \to a$$

- Input: board state
- Output: move/action
- Methods: search, planning, reinforcement learning

Clustering: Finding Groups



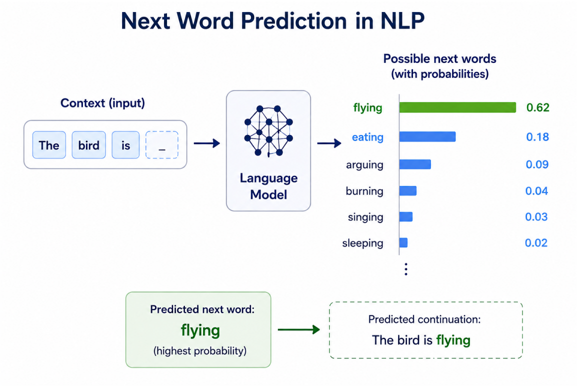
Task

Group similar data points without predefined labels.

$$x_i \rightarrow \text{cluster}_k$$

- Output: groups
- Example: customer segmentation
- Learning type: unsupervised learning

Natural Language Processing



Task

Process, generate, or transform human language.

$$f_\theta(\text{text}) \to \text{text}$$

- Translation
- Summarization
- Question answering
- Chatbots

Main Limitations

V. Main Limitations

Audience question

We know AI is powerful, but what are the limitations ?

Main Limitations of AI

Limitation	Consequence
Data dependency	No data, no learning
Bias in data	Biased model behavior
Generalization	Failure on unseen situations
Interpretability	Difficult to explain decisions
Robustness	Sensitivity to noise or distribution shift
Computational cost	Expensive training and deployment
Ethics and regulation	Privacy, responsibility, fairness

Key message

AI systems do not understand the world in the human sense. They learn patterns from data.

Performing a Cognitive Task

VI. Performing a Cognitive Task

Audience question

What does it mean mathematically today?

Performing a Cognitive Task

Mathematical view

Many AI tasks can be written as learning a function:

$$f_\theta(x) \approx y$$

where x is the input, y is the expected output, and θ are the model parameters.

- The model receives an input x.
- It produces a prediction:

$$\hat{y} = f_\theta(x)$$

- The prediction is compared to the expected output y.
- The difference is measured by a loss function:

$$\mathcal{L}(y, \hat{y})$$

- Learning means updating θ to reduce this loss.

Linear Regression

Linear regression is one of the simplest machine learning models.
It predicts an output $\hat{y}$ from an input x using a straight line:

$$\hat{y} = wx + b$$

- w is the **weight**: it controls the slope of the line.
- b is the **bias**: it shifts the line up or down.

Key idea

The values of w and b are the parameters learned during training.

Learning Process: Step by Step

1. **Input:** give the model an example x.
2. **Prediction:** the model computes:

$$\hat{y} = f_\theta(x)$$

3. **Error measurement:** compare the prediction $\hat{y}$ with the true answer y:

$$\mathcal{L}(y, \hat{y})$$

4. **Parameter update:** modify θ to reduce the error.
5. **Repeat:** do this many times on many examples.

Linear Regression: Learning Objective

Mean Squared Error

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$$

Important

This is a
Loss Function.

The optimizer will
minimize this quantity.

- If the prediction is close to the target, the error is small.
- If the prediction is far from the target, the error is large.
- The model learns by finding w and b that minimize the MSE.

$$(w^\star, b^\star) = \arg\min_{w,b} MSE$$

Properties of a Good Loss Function

- **Non-negative:**

$$\mathcal{L}(y, \hat{y}) \geq 0$$

- **Zero for perfect prediction:**

$$\mathcal{L}(y, \hat{y}) = 0 \iff y = \hat{y}$$

- **Differentiable:** enables gradient descent.
- **Stable gradients:** avoids exploding or vanishing updates.
- **Convexity:** easier optimization, but not mandatory.
- **Task-aligned:** reflects the real objective.

Convexity

Every local minimum is also a global minimum.

This is useful, but optional. Deep learning losses are often non-convex.

Gradient Descent: How the Model Learns

Idea

Gradient descent updates the parameters in the direction that reduces the error.

$$w \leftarrow w - \eta \frac{\partial \mathcal{L}}{\partial w}$$

$$b \leftarrow b - \eta \frac{\partial \mathcal{L}}{\partial b}$$

- η: learning rate.
- $\frac{\partial \mathcal{L}}{\partial w}$: tells how the loss changes when w changes.
- $\frac{\partial \mathcal{L}}{\partial b}$: tells how the loss changes when b changes.
- The process is repeated until the error becomes small.

Linear Regression Optimization

Model prediction

$$\hat{y}_i = wx_i + b$$

Mean Squared Error (MSE)

$$\mathcal{L}(w, b) = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2$$

Gradient with respect to w

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial w} &= \frac{\partial}{\partial w} \left(\frac{1}{N} \sum_{i=1}^{N} (y_i - (wx_i + b))^2 \right) \\
&= \frac{1}{N} \sum_{i=1}^{N} 2(y_i - \hat{y}_i)(-x_i) \\
&= -\frac{2}{N} \sum_{i=1}^{N} x_i (y_i - \hat{y}_i) = -\frac{2}{N} \sum_{i=1}^{N} x_i e_i
\end{aligned}$$

Gradient Descent Update

Gradient with respect to b

$$\frac{\partial \mathcal{L}}{\partial b} = \frac{\partial}{\partial b}\left(\frac{1}{N}\sum_{i=1}^{N}(y_i - (wx_i + b))^2\right)$$
$$= \frac{1}{N}\sum_{i=1}^{N} 2(y_i - \hat{y}_i)(-1)$$
$$= -\frac{2}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i) = -\frac{2}{N}\sum_{i=1}^{N} e_i$$

Gradient descent update

$$w \leftarrow w - \eta\frac{\partial \mathcal{L}}{\partial w}, \qquad b \leftarrow b - \eta\frac{\partial \mathcal{L}}{\partial b}$$

Gradient sign

$$\frac{\partial \mathcal{L}}{\partial w} > 0 \Rightarrow \mathcal{L}(w) \text{ increases with } w, \qquad \frac{\partial \mathcal{L}}{\partial w} < 0 \Rightarrow \mathcal{L}(w) \text{ decreases with } w.$$

What You Should Remember

VII. What You Should Remember

1. AI is not magic.
2. AI is a broad field, not only deep learning.
3. Most modern AI systems learn from data.
4. Learning often means fitting a function.
5. The error tells the model what to improve.
6. The quality of the data strongly affects the quality of the model.

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